

JOSHITHA RAJASEKARAN

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LinkedIn | GitHub | Publication

PROFESSIONAL SUMMARY

Motivated Computer Science and Engineering graduate with hands-on experience in **software development, back-end application development, AWS cloud services, and AI-based systems**. Proficient in **Java, Python, C, Data Structures, Algorithms, DBMS, Operating Systems, and Computer Networks**. Built and deployed projects involving **Flask, TensorFlow, OpenCV, and cloud infrastructure**, with a strong focus on **problem solving, scalable application development, debugging, and writing clean code**. Eager to contribute as a **Software Development Engineer** in fast-paced engineering environments and build reliable products that deliver impact at scale.

TECHNICAL SKILLS

Programming Languages	Java, Python, C
Core Computer Science	Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
Backend / Development	Flask, REST-based application development, Web Application Development
Cloud / Dev Tools	AWS EC2, AWS S3, IAM, CloudWatch, Git, GitHub, VS Code
Libraries / Technologies	TensorFlow, OpenCV, NumPy, Pandas, SciPy, Librosa
Web Technologies	HTML5, CSS3, JavaScript

INTERNSHIP EXPERIENCE

Cloud Computing Intern <i>Orionshift, Chennai</i>	Jul 2025 – Aug 2025
- Managed and supported AWS EC2 instances, S3 buckets, IAM policies, and CloudWatch monitoring for cloud-based environments, strengthening practical knowledge of deployment and infrastructure operations. - Assisted in troubleshooting cloud configuration and service issues to improve system reliability and maintain service uptime. - Created infrastructure documentation and setup guidelines, reducing environment setup effort by 30% and improving onboarding efficiency. - Worked on resource monitoring, access management, and operational support tasks, building familiarity with scalable cloud systems and production workflows.	
Full Stack Web Development Intern <i>Skill Intern</i>	2024
- Developed responsive web application components using HTML, CSS, and JavaScript, with focus on usability, layout consistency, and clean implementation. - Worked on frontend-backend integration concepts and gained exposure to application flow, debugging, and deployment-oriented development practices. - Used Git and GitHub for version control, source code management, and iterative development.	

PROJECTS

Speech-to-Text Transcription System	Python, TensorFlow, Flask, Librosa, SciPy
- Built an AI-powered speech-to-text system using Python, TensorFlow, Librosa, and SciPy to process and transcribe audio inputs. - Improved transcription accuracy by 18% through audio preprocessing and noise augmentation techniques. - Processed and analyzed 500+ audio samples using MFCC and spectrogram-based feature extraction to improve model performance. - Developed a Flask-based web application for real-time transcription, demonstrating backend application development and integration of ML models into usable software workflows.	
Smart Accident Detection and Emergency Human Injury Alert System	Python, TensorFlow, OpenCV, NumPy
Designed and implemented a computer vision-based accident detection system using TensorFlow, OpenCV, and NumPy.	

- Built a CNN-based model that achieved **92% detection accuracy** on **1000+ labeled images**.
- Integrated accident severity identification and emergency alert workflow concepts to simulate real-time response functionality.
- Strengthened skills in **problem solving, model development, data preprocessing, and system-oriented application design**.

PUBLICATION

Smart Accident Detection and Emergency Response System with Accident Severity Classification
International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), Vol. 6, Issue 1, May 2026

Authors: S. Madhan, Joshitha R, Pravina M, Nithyasri K

[View Publication](#)

EDUCATION

B.E. in Computer Science and Engineering

University College of Engineering, Thirukkuvai

2022 – 2026

CGPA: 8.3 / 10.0

CERTIFICATIONS

- Cloud Computing Internship – Orionshift
- Full Stack Web Development – Skill Intern
- Web Development Internship Certificate
- Academic Rank Holder – Computer Science Engineering